



Biswajit Jana <biswajitj998@gmail.com>

Stage 2: Fwd: Long-run feedback experiment (Exp-H) — results and discussion

2 messages

Biswajit Jana <biswajitj998@gmail.com>

25 February 2026 at 11:34

To: William Martin <w.e.martin@herts.ac.uk>

Cc: Supriyo Ghosh <s.ghosh3@herts.ac.uk>, Hugh Jones <h.r.a.jones@herts.ac.uk>

Here is the string of the email and the numbers below for Stage 2 discussion

----- Forwarded message -----

From: Biswajit Jana <biswajitj998@gmail.com>

Date: Fri, 2 Jan 2026, 11:32

Subject: Re: Long-run feedback experiment (Exp-H) — results and discussion

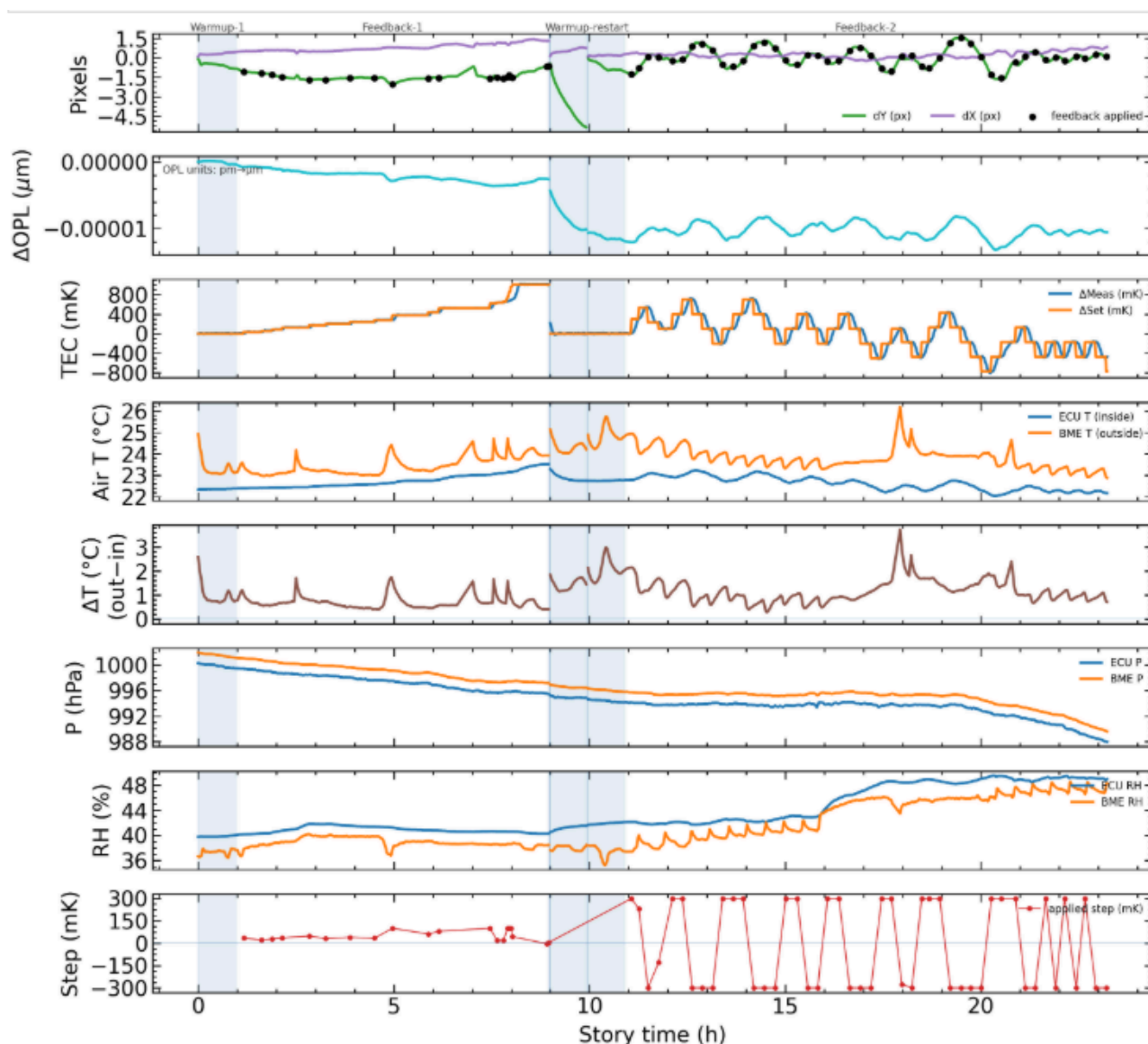
To: William Martin <w.e.martin@herts.ac.uk>

Cc: Supriyo Ghosh <s.ghosh3@herts.ac.uk>, Hugh Jones <h.r.a.jones@herts.ac.uk>, Krupa Kuruvila <k.m.kuruvila@herts.ac.uk>

Morning everyone,

Happy New Year — I hope you all had a restful break. I didn't want to message during the holidays, but I wanted to share the results from my most recent long-run hybrid OPL $\oplus$ PX feedback experiment, which I think are now in a good state for discussion. The summary plots and console statistics are attached.

Experiment overview

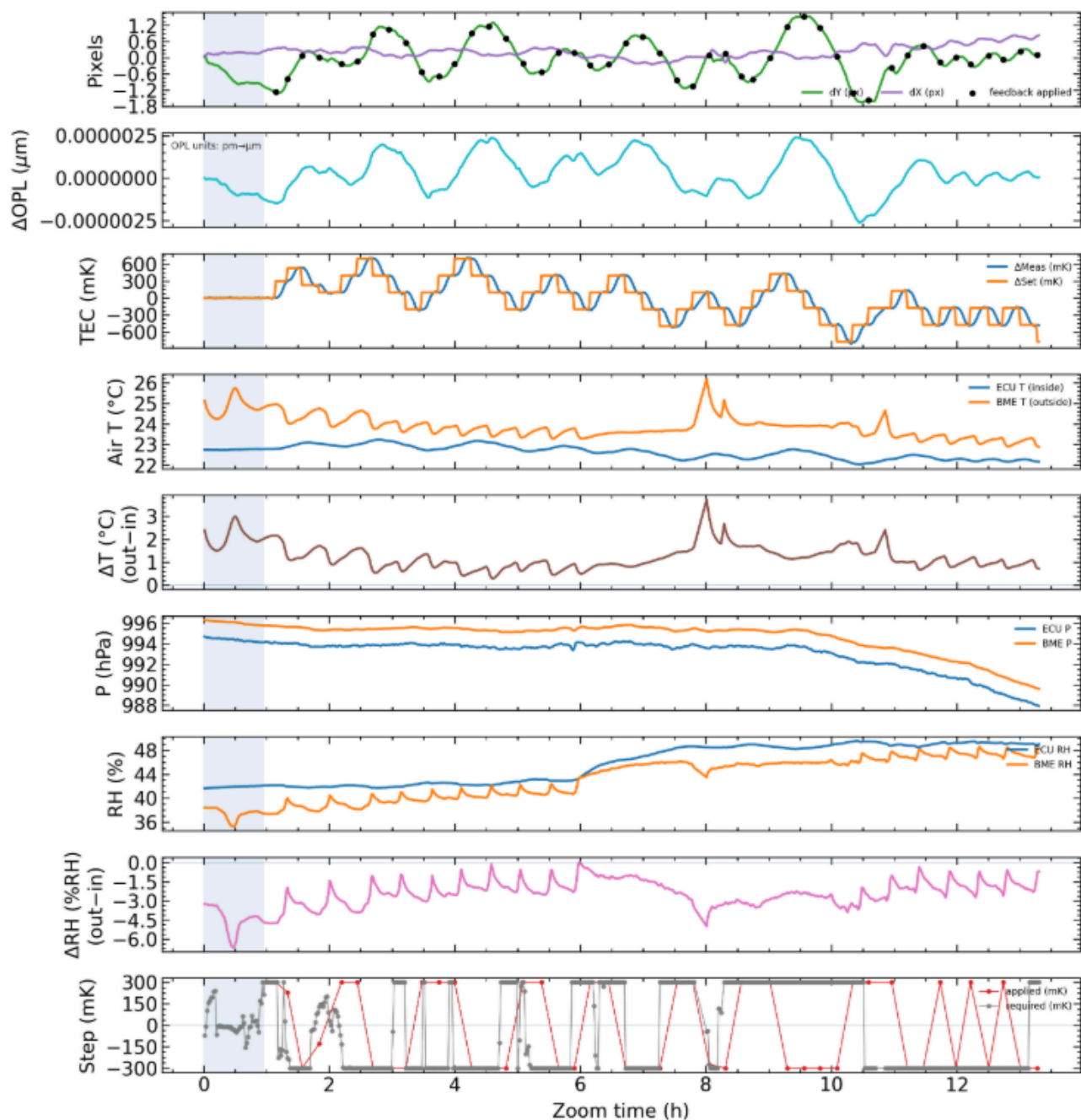


I stitched three sequential runs into a single continuous "story-time" timeline: an adaptive afternoon run, a short warmup-only restart, and a long overnight feedback run. (Above is the plot for it) The stitched view helps show the full environmental evolution, while the zoomed

plot focuses on the best continuous feedback performance.

- **Time span (stitched):** 2025-12-17 12:55 → 2025-12-18 12:59
- **Total duration:** 24.1 h
- **Median cadence:** 1.25 min
- **Segments:** Warmup-1 → Feedback-1 → Warmup-restart → Feedback-2
- **Overall feedback duty cycle:** 6.2%

The initial Warmup-1 → Feedback-1 segment did not perform reliably. Around ~22:00 I stopped the experiment remotely after identifying a control-logic issue, fixed the code, and restarted shortly afterwards. The **Warmup-restart and Feedback-2** segment therefore represents the cleanest and most representative dataset: ~14 h usable data, including **12.3 h of sustained closed-loop feedback**. **ZOOMED OUT VERSION BELOW:**



Main quantitative result (Warmup-restart → Feedback-2)

- $\sigma(dY)$: reduced by **60.4%** ($1.751 \rightarrow 0.693$ px)
- $RMS(dY)$: reduced by **74.5%** ($2.721 \rightarrow 0.693$ px)
- $\sigma(\Delta OPL)$: reduced by **45.8%** ($1.99 \times 10^{-6} \rightarrow 1.08 \times 10^{-6}$ μm)
- **Time within ± 0.10 px:** 16.2% (± 0.20 px: 29.5%)
- **Long-term drift during Feedback-2:** ≈ 0 px h^{-1} (linear slope -0.001 px h^{-1})

The near-zero drift rate indicates that, although short-timescale oscillations remain, the feedback is effectively suppressing **secular drift over multi-hour timescales**.

Environmental behaviour

During Feedback-2:

- **ΔT(out-in):** 3.45 °C span
- **ΔRH(out-in):** 5.1 %RH span
- **ΔP(out-in):** 0.63 hPa span

After ~19:00 the building AC appears to switch off. The room AC remained on, so the inside temperature stayed comparatively stable while the outside sensor continued to drift and cycle. The residual internal ripples are consistent with coupling through enclosure leakage and thermal inertia rather than direct setpoint changes. More details can be shared in the meeting. Lag analysis suggests pixel motion responds to temperature-related quantities on **~5–15 min timescales**, consistent with the system's combined thermal and mechanical response.

Control authority

Empirically, the applied steps correspond to $\sim 5.6 \times 10^{-5} \text{ px mK}^{-1}$, so even at the $\pm 300 \text{ mK}$ clamp the immediate next-sample correction is only $\sim 0.02 \text{ px}$. This confirms the controller works via **incremental correction over many cycles**, not single-step cancellation, which matches the observed lag behaviour.

Remote access update

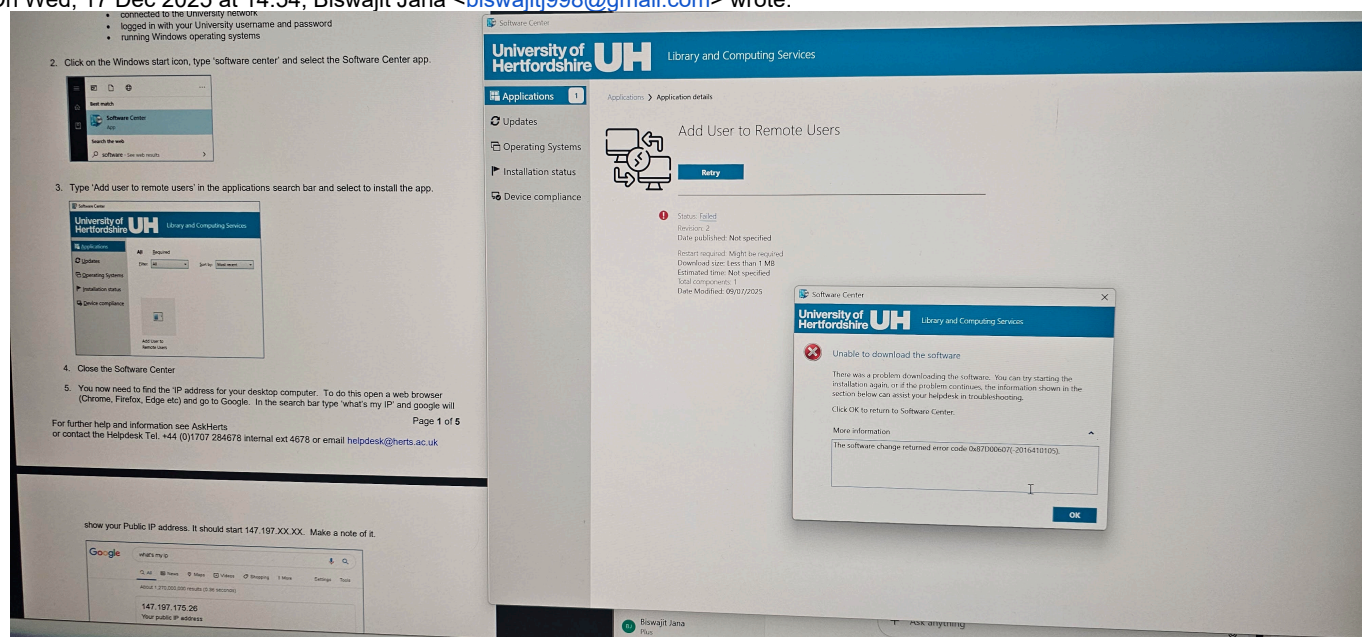
On a practical note, I've tested **Chrome Remote Desktop** and it works very well for long runs. As long as my email remains active on the control PC, I can monitor and control the system from anywhere. It's free, open-source, and avoids the need for AnyDesk or the university remote desktop tools. This should be sufficient for monitoring, stopping runs, and safe shutdowns; physical power-cycling would still require on-site access.

If possible, could we have a short meeting or discussion on **Monday or Wednesday next week**, depending on availability (Bill, Supriyo, Hugh), to review these results and agree on next steps? Also, **Krupa**, could you let me know when the labs officially reopen? Thanks very much, any follow up questions before the meeting please ask and here's hoping 2026 brings a bit more **stability and precision** to our little EXOhSPEC 😊

Best Regards,
Biswajit

PS: Need to check units of OPL

On Wed, 17 Dec 2025 at 14:54, Biswajit Jana <biswajit998@gmail.com> wrote:



Installation failure. Don't know what's the reason. Can reach out to helpdesk. But will leave this as it is for now until next year

On Mon, 15 Dec 2025, 14:41 William Martin, <w.e.martin@herts.ac.uk> wrote:

Sounds good.

Have a look at this:

[herts365.sharepoint.com/sites/Computing/SiteAssets/Forms/AllItems.aspx?id=%2Fsites%2FComputing%2FSiteAssets%2FSitePages%2FGet-connected%2FRemote access to university desktop computer%2Epdf&parent=%2Fsites%2FComputing%2FSiteAssets%2FSitePages%2FGet-connected](https://herts365.sharepoint.com/sites/Computing/SiteAssets/Forms/AllItems.aspx?id=%2Fsites%2FComputing%2FSiteAssets%2FSitePages%2FGet-connected%2FRemote%20access%20to%20university%20desktop%20computer%2Epdf&parent=%2Fsites%2FComputing%2FSiteAssets%2FSitePages%2FGet-connected)

Bill

From: Biswajit Jana <biswajit998@gmail.com>

Sent: Monday, December 15, 2025 13:17

To: William Martin <w.e.martin@herts.ac.uk>

Cc: Krupa Kuruvila <k.m.kuruvila@herts.ac.uk>; Hugh Jones <h.r.a.jones@herts.ac.uk>; Supriyo Ghosh <s.ghosh3@herts.ac.uk>

Subject: Re: Long-run feedback experiment (Exp-H) — results and discussion

Hi Bill,

Just had conversation with Luke, he won't be coming on Wednesday so I'll be there bit earlier than my usual time.

Regards,
Biswajit

On Mon, 15 Dec 2025, 10:20 Biswajit Jana, <biswajit998@gmail.com> wrote:

Hi Bill,

Are you in Spectra today? The Uni feels a bit like a low-density region of the Universe this morning — not many people around.

 **Experiment console report.txt**
7K

Biswajit Jana <biswajit998@gmail.com>
Draft To: Supriyo Ghosh <s.ghosh3@herts.ac.uk>

26 March 2026 at 22:58

 **Experiment console report.txt**
7K